Stochastic Phase Uncertainty in Macromolecular Crystallography

State of Knowledge, Comparative Framework, Mathematical Foundations, and Benchmark Proofs

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This document consolidates the complete theoretical and computational framework connecting the EDENN.waters stochastic omit ensemble to classical Blow–Crick maps, Sigma-A estimation, Lunin–Skovoroda LB estimates, PanDDA, composite OMIT maps, qFit, and polder maps. It establishes formal propositions with benchmark proofs for each principal claim.

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1 The Phase Problem: Mathematical Foundations

1.1 The crystallographic Fourier transform

The electron density at position $\mathbf{r}$ in the unit cell is related to the complex structure factors $F(h)$ by:

$$\rho(\mathbf{r}) = \frac{1}{V} \sum_h F(h)\, e^{-2\pi i\, \mathbf{h}\cdot\mathbf{r}} \tag{1}$$

where V is the unit-cell volume and the sum runs over all Miller indices $\mathbf{h} = (h, k, l)$. Each structure factor decomposes as $F(h) = |F(h)|\, e^{i\varphi(h)}$.

From a diffraction experiment one measures intensities $I(h) \propto |F(h)|^2$, yielding amplitudes $|F_{\text{obs}}(h)|$ but *no phase information whatsoever*. This is the phase problem. Without phases, equation (1) cannot be evaluated, and no electron density map can be computed.

1.2 Why phases dominate structural information

Shannon's theorem implies that for a structure with N_{atoms} atoms the phase of $F(h)$ carries approximately twice as much information as the amplitude. Empirically, swapping the phases of two unrelated proteins while keeping each other's amplitudes produces a density map that resembles the protein whose phases were used, not the protein whose amplitudes were used. Amplitudes constrain the map; phases determine it.

1.3 The phase probability distribution

Because individual phases are unmeasured, the framework of probabilistic crystallography treats φ_h as a random variable with a probability distribution $P(\varphi_h)$ that encodes all available information.

Definition 1.1: Phase probability distribution

The phase probability distribution $P(\varphi_h \mid \text{data}, \text{model})$ is the posterior distribution of the phase of reflection h given the observed amplitude $|F_{\text{obs}}(h)|$ and the current structural model. It is a circular probability distribution supported on $(-\pi, \pi]$.

Under most models, $P(\varphi_h)$ takes the von Mises (circular normal) form:

$$P(\varphi_h) = \frac{1}{2\pi\, I_0(X_h)} \exp\!\big[X_h \cos(\varphi_h - \varphi_{\text{mod}})\big] \tag{2}$$

where φ_{mod} is the model phase, $X_h \geq 0$ is the concentration parameter, and I_0 is the modified Bessel function of order 0.

2 Blow–Crick Best Maps (1959)

2.1 Derivation

Theorem 2.1. Blow–Crick optimality

The density estimator $\hat{\rho}(\mathbf{r})$ that minimises the expected mean-square error $\mathbb{E}\big[\|\hat{\rho} - \rho_{\text{true}}\|^2\big]$ over all linear estimators of the form $\hat{\rho} = \mathcal{F}^{-1}[w_h F_{\text{obs}} e^{i\varphi_{\text{mod}}}]$ has weights $w_h = m_h$, giving:

$$\rho_{\text{best}}(\mathbf{r}) = \mathcal{F}^{-1}\big[m_h\, |F_{\text{obs}}|\, e^{i\varphi_{\text{mod}}}\big] \tag{3}$$

where the *figure of merit* is $m_h \equiv \mathbb{E}[\cos(\varphi_h - \varphi_{\text{mod}})] = I_1(X_h)/I_0(X_h)$.

Proof. Write the true density as $\rho_{\text{true}} = \mathcal{F}^{-1}[F_{\text{true}} e^{i\varphi_{\text{true}}}]$. The mean-square error in Fourier space is $\mathbb{E}\big[|w_h F_{\text{obs}} e^{i\varphi_{\text{mod}}} - F_{\text{true}} e^{i\varphi_{\text{true}}}|^2\big]$. Differentiating with respect to w_h and setting to zero yields $w_h = \mathbb{E}[\cos(\varphi_h - \varphi_{\text{mod}})]$, which is the definition of m_h. □

2.2 Geometric interpretation

The FOM m_h is the magnitude of the centroid of the phase probability distribution on the unit circle:

$$m_h = \big|\mathbb{E}[e^{i\varphi_h}]\big| = \left|\int_{-\pi}^{\pi} e^{i\varphi_h} P(\varphi_h)\, d\varphi_h\right| \tag{4}$$

The centroid lies *inside* the Argand circle of radius $|F_{\text{obs}}|$, at distance $m_h|F_{\text{obs}}|$ from the origin. As phase uncertainty increases, the centroid moves toward the origin and $m_h \to 0$. For a perfectly known phase, the centroid lies on the circle and $m_h = 1$.

2.3 The FOM–concentration relationship

For the von Mises distribution (2):

$$m_h = \frac{I_1(X_h)}{I_0(X_h)} \tag{5}$$

This is a monotone increasing bijection from $X_h \geq 0$ to $m_h \in [0, 1)$. Inversion to obtain X_h from m_h is performed by Newton–Raphson on $f(X_h) = I_1(X_h)/I_0(X_h) - m_h = 0$, using the Mardia approximation $X_h \approx m_h(3 - m_h^2)/(1 - m_h^2)$ as the starting point.

2.4 The circular problem: how to compute m_h without φ_{true}

The definition of m_h involves $\mathbb{E}[\cos(\varphi_h - \varphi_{\text{true}})]$, which requires knowledge of φ_{true} — the very quantity we cannot observe. In practice m_h is **never computed from the true phase**; it is a property of the probability distribution $P(\varphi_h)$ alone.

The identity $m_h = I_1(X_h)/I_0(X_h)$ makes this explicit: given the concentration X_h (which is derived from the model and the data via a likelihood function), m_h follows algebraically. Different methods for estimating X_h are what distinguish the competing methods reviewed below.

3 Sigma-A Estimation (Read, 1986)

3.1 The coordinate error model

Read (1986) formalised Luzzati's coordinate error model. For a model with root-mean-square coordinate error σ_x, the expected relationship between model and true structure factors is:

$$\langle |F_{\text{mod}}(h)| \rangle = D_h\, |F_{\text{true}}(h)|, \qquad D_h = \exp(-2\pi^2 \sigma_x^2 s^2) \tag{6}$$

where $s = |\mathbf{h}|/2$ is the scattering vector magnitude ($s^2 = \sin^2\theta/\lambda^2$). The remainder $\varepsilon_h = F_{\text{true}} - D_h F_{\text{mod}}$ is modelled as a random variable with expected squared modulus $\Sigma_h = \sigma_\varepsilon^2 = \langle |\varepsilon_h|^2 \rangle$.

Under Wilson statistics, the full likelihood of observing F_{obs} given the model takes the Rice distribution form. The resulting phase probability is von Mises with concentration:

$$X_h^{\Sigma A} = \frac{2\, D_h\, |F_{\text{obs}}||F_{\text{mod}}|}{\Sigma_h} \tag{7}$$

3.2 Estimation of D_h and Σ_h

Both parameters are estimated by maximum likelihood from the *working set* of reflections in resolution shells:

$$\hat{D}_k = \arg\max_{D} \prod_{h \in \text{shell}_k} \mathcal{L}(F_{\text{obs}}(h) \mid D, \Sigma) \tag{8}$$

$$\hat{\Sigma}_k = (\text{jointly estimated with } D_k) \tag{9}$$

This is the approach used in REFMAC5 (Murshudov et al. 1997) and Phenix (Adams et al. 2010).

3.3 The working-set bias problem

Proposition 3.1: Working-set FOM inflation

Sigma-A FOM estimates computed from the working set of a refined model systematically overestimate m_h, producing a map whose apparent phase quality exceeds the true phase quality.

Argument. Refinement minimises a target function that is equivalent to maximising the working-set likelihood. The maximum likelihood estimate $\hat{D}_k$, $\hat{\Sigma}_k$ therefore reflects not only the true structural accuracy but also the fit of the model to the working amplitudes. For a model refined with many degrees of freedom relative to the number of reflections, this overfitting can be severe. The result is $\hat{D}_k > D_k^{\text{true}}$ and $\hat{\Sigma}_k < \Sigma_k^{\text{true}}$, inflating X_h and hence m_h via equation (7). In the limit of an exactly fitted model, $\hat{\Sigma}_k \to 0$, $X_h \to \infty$, and $m_h \to 1$ for all reflections regardless of the true phase accuracy.

4 Lunin–Skovoroda LB Estimates (1995)

4.1 Motivation

Lunin & Skovoroda (1995) identified the working-set bias as the principal limitation of Sigma-A maps and proposed to estimate D_h and Σ_h exclusively from the *free* reflections — those excluded from refinement by the Brünger R-free protocol (Brünger 1992).

4.2 The LB formula

The phase probability is parameterised as:

$$P(\varphi_h) \propto \exp\left[\alpha_k |F_{\text{obs}}||F_{\text{mod}}| \cos(\varphi_h - \varphi_{\text{mod}}) + \beta_k |F_{\text{obs}}|^2\right] \tag{10}$$

for reflection h in resolution shell k. The parameters α_k, β_k are estimated by maximising the likelihood over the R-free test set:

$$(\hat{\alpha}_k, \hat{\beta}_k) = \arg\max_{\alpha,\beta} \prod_{h \in \text{R-free}_k} P(F_{\text{obs}}(h) \mid \alpha, \beta) \tag{11}$$

The connection to Sigma-A is: $\alpha_k = 2D_k/\Sigma_k^{\text{free}}$ and $\beta_k = -1/\Sigma_k^{\text{free}}$, so the LB estimate recovers the same functional form but with parameters conditioned on test-set observations only.

4.3 Why LB estimates are unbiased

Because the free reflections were never used in refinement, the model has no systematic bias toward matching their amplitudes. The maximum likelihood estimate therefore reflects the genuine structural accuracy, not the refinement fit. Lunin & Skovoroda demonstrated in Figure 1 of their 1995 paper that LB estimates agree with the true phase error (measured from known structures) far more accurately than working-set Sigma-A estimates at equivalent resolutions.

4.4 Limitation

LB estimates are a single point estimate of (α_k, β_k) per shell — they do not provide per-reflection FOM values or any spatial information about which *regions* of the structure have high or low phase quality.

5 Composite OMIT Maps (Phenix)

5.1 The model bias problem

The $2F_o - F_c$ map computed from a refined model is biased toward the model by construction: the phases φ_{mod} pull the density toward the model's atomic positions whether or not those positions are correct. This model bias is particularly severe for:

- Incorrectly placed atoms
- Unmodelled features (ligands, waters, conformational variants)
- Regions with coordinate error exceeding $\sim$1 Å

5.2 The OMIT strategy

The classical solution is to omit a region of interest from the model, refine the remaining atoms, and compute phases from the omit model. The map in the omitted region is then uncontaminated by the phases of the omitted atoms. The limitation is that omitting a large region degrades the phases everywhere due to the reduced completeness of the model.

Phenix composite OMIT maps (Terwilliger et al. 2008) resolve this by:

1. Partitioning the unit cell into overlapping sub-volumes
2. Simulating the omit of each sub-volume via simulated-annealing refinement (which introduces random perturbations that average out model bias in that region)
3. Tiling the resulting maps into a composite whole-cell map

The composite map thus has reduced model bias everywhere simultaneously.

5.3 Mathematical character

The composite OMIT map is not a statistically optimal estimator of ρ_{true}. It is an ad hoc debiasing procedure whose statistical properties are not rigorously characterised. No formal connection exists between the composite OMIT map and the Blow-Crick optimal map, nor does it produce FOM estimates or phase distributions.

6 PanDDA (Pearce et al., 2017)

6.1 Motivation: the fragment screening problem

In fragment-based drug discovery, hundreds to thousands of datasets are collected from crystals soaked with small molecules. Binding events are rare (5–30% hit rate) and typically occur at low occupancy (10–40%), making them invisible in standard $2F_o - F_c$ maps.

6.2 The PanDDA algorithm

Ground state estimation. For each dataset d in a campaign of N datasets, the ground state is estimated as the per-reflection mean amplitude across all datasets after resolution-shell scaling:

$$\langle |F_{\text{obs}}(h)| \rangle_{\text{GS}} = \frac{1}{N} \sum_{d=1}^{N} s_d \, |F_{\text{obs}}^{(d)}(h)| \tag{12}$$

where s_d are per-dataset scale factors. For a dataset containing a bound ligand, $|F_{\text{obs}}^{(d)}|$ is systematically elevated at ligand-related reflections; however, if most datasets are apo, the mean is dominated by the apo signal.

Z-score per reflection.

$$Z^{(d)}(h) = \frac{s_d |F_{\text{obs}}^{(d)}(h)| - \langle |F_{\text{obs}}| \rangle}{\sigma_d(h)} \tag{13}$$

where $\sigma_d(h)$ is the inter-dataset standard deviation. Reflections with $Z^{(d)}(h) > 3$ are flagged as carrying bound-state signal in dataset d.

Event map. The BDC-corrected event map is:

$$\rho_{\text{event}}^{(d)}(\mathbf{r}) = \rho_{\text{obs}}^{(d)}(\mathbf{r}) - (1 - \text{BDC}) \, \rho_{\text{GS}}(\mathbf{r}) \tag{14}$$

where BDC (Background Dataset Contribution) is estimated as the fraction of the observed density explained by the ground state outside the event region.

6.3 Statistical properties of PanDDA

- Sensitivity scales as q/σ_N where $\sigma_N \propto 1/\sqrt{N}$
- For $N = 100$ datasets, sensitivity to $q \sim 5\%$ occupancy is achievable
- The method makes no assumptions about the bound-state model — it requires only an apo model for initial phase estimation
- FOM estimation is implicit (not computed) — the method operates purely on amplitude statistics

6.4 Fundamental requirement and limitation

PanDDA *requires* a screening campaign. Its ground state estimation is only reliable when $N \gg 1/(1 - f_{\text{hit}})$, i.e. when the majority of datasets are ligand-free and the mean amplitude is dominated by the apo protein. For a single dataset, equation (12) reduces to the single observed amplitude, and the Z-score becomes undefined.

7 Polder Maps (Liebschner et al., 2017)

Polder maps address a specific failure mode of standard OMIT maps near the protein surface: when a small molecule is omitted from the model, the bulk-solvent correction floods the omit region with a flat featureless density, masking the signal from the missing atom.

The polder algorithm excludes the bulk-solvent mask from the omit region, preventing this flooding. The resulting map amplifies weak density near the protein surface, making it sensitive to partially occupied solvent molecules and small ligands in shallow pockets.

Polder maps produce no FOM estimates and no formal statistical framework. They are a targeted debiasing method for a specific geometric context — ordered solvent near the protein surface — and are not applicable to buried features or features in the bulk solvent region.

8 qFit (van den Bedem et al., 2009; Riley et al., 2021)

qFit builds multiconformer models from single-crystal X-ray or cryo-EM data by sampling rotamers and backbone conformations, then selecting the combination that best explains the observed density via a regularised linear program:

$$\min_{\mathbf{q}\geq 0,\, \mathbf{1}^T\mathbf{q}=1} \|\boldsymbol{\rho}_{\text{obs}} - \boldsymbol{A}\mathbf{q}\|^2 + \lambda\|\mathbf{q}\|_1 \tag{15}$$

where $\boldsymbol{A}$ is a dictionary of rotamer density contributions and $\mathbf{q}$ are the conformer occupancies.

qFit operates entirely in real space on a precomputed density map — it does not propagate phase uncertainty, does not estimate FOMs, and does not produce event maps. Its relationship to EDENN.autobuild is that qFit can use the density ensemble produced by the EDENN.waters sampling as its input map, allowing phase uncertainty to flow into the conformational ensemble.

9 The EDENN.waters Framework: Stochastic Omit Ensemble

9.1 The stochastic omit ensemble

The EDENN.waters protocol generates an ensemble of K electron density maps by:

1. Randomly selecting a fraction P_{omit} of atoms from the model
2. Computing phases from the remaining atoms
3. Calculating the map $\rho^{(k)}(\mathbf{r}) = \mathcal{F}^{-1}[|F_{\text{obs}}|\, e^{i\varphi^{(k)}}]$
4. Repeating K times

The fundamental observable is the **ensemble mean**:

$$\rho_{\text{ens}}(\mathbf{r}) = \frac{1}{K}\sum_{k=1}^{K} \rho^{(k)}(\mathbf{r}) \tag{16}$$

and the **SNR map**:

$$\text{SNR}(\mathbf{r}) = \frac{\langle \rho^{\text{omit}} \rangle(\mathbf{r})}{\text{std}[\rho^{\text{omit}}](\mathbf{r})} \tag{17}$$

9.2 The central identity

Proposition 9.1: Mean map identity

In the limit $K \to \infty$, the ensemble mean map equals the Blow–Crick best map:

$$\lim_{K\to\infty} \rho_{\text{ens}}(\mathbf{r}) = \mathcal{F}^{-1}\big[m_h\, |F_{\text{obs}}|\, e^{i\varphi_{\text{mod}}}\big] = \rho_{\text{best}}(\mathbf{r}) \tag{18}$$

where $m_h = \mathbb{E}[\cos(\delta\varphi_h)]$ and $\delta\varphi_h$ is the phase perturbation induced by the omit fraction.

Proof. By the law of large numbers, as $K \to \infty$:

$$\begin{aligned} \rho_{\text{ens}}(\mathbf{r}) &\to \mathbb{E}\big[\rho^{\text{omit}}(\mathbf{r})\big] \\ &= \mathcal{F}^{-1}\big[\mathbb{E}\big[|F_{\text{obs}}|\, e^{i\varphi^{\text{omit}}}\big]\big] \\ &= \mathcal{F}^{-1}\big[|F_{\text{obs}}|\, \mathbb{E}\big[e^{i\varphi^{\text{omit}}}\big]\big] \end{aligned} \tag{19}$$

The expectation $\mathbb{E}[e^{i\varphi^{\text{omit}}}]$ is the characteristic function of the phase distribution at argument 1. For the von Mises distribution (2):

$$\mathbb{E}[e^{i\varphi^{\text{omit}}}] = m_h\, e^{i\varphi_{\text{mod}}} \tag{20}$$

because the characteristic function of vonMises(μ, κ) at integer argument n is $I_n(\kappa)/I_0(\kappa)\, e^{in\mu}$, giving at $n = 1$: $I_1(X_h)/I_0(X_h)\, e^{i\varphi_{\text{mod}}} = m_h\, e^{i\varphi_{\text{mod}}}$. Substituting and applying $\mathcal{F}^{-1}$ gives the Blow–Crick formula (3). □ □

9.3 FOM estimation from the ensemble

Proposition 9.2: Reciprocal-space FOM estimator

The per-reflection figure of merit can be estimated directly from the ensemble as:

$$\hat{m}_h = \frac{|\langle \tilde{F}_h^{\text{omit}} \rangle|}{|F_{\text{obs}}(h)|} \tag{21}$$

where $\langle \tilde{F}_h^{\text{omit}} \rangle = \mathcal{F}[\rho_{\text{ens}}]_h$ is the Fourier coefficient of the mean map at reflection h.

Proof. From the mean-map identity, $\mathcal{F}[\rho_{\text{ens}}]_h = m_h |F_{\text{obs}}| e^{i\varphi_{\text{mod}}}$ in the large-K limit. Taking the modulus and dividing by $|F_{\text{obs}}|$ yields (21). □ □

Bias property. Because the omit perturbations are not conditioned on fitting any specific subset of reflections, $\hat{m}_h$ does not inherit the working-set inflation that afflicts Sigma-A estimates. It is an unbiased estimator of the true FOM in the same sense that R-free is an unbiased estimator of the true R-factor.

9.4 Real-space SNR to FOM conversion

For a candidate feature at position $\mathbf{r}$ dominated by a single reflection h (the isolated-site approximation):

$$\text{SNR}(\mathbf{r}) \approx m_h \sqrt{\frac{2}{1 - m_h^2}} \tag{22}$$

Inverting:

$$\hat{m}(\mathbf{r}) = \frac{\text{SNR}}{\sqrt{2 + \text{SNR}^2}} \tag{23}$$

This provides a real-space route to FOM estimation that is completely independent of the reciprocal-space estimator (21), enabling the cross-consistency check described in Section 13.

9.5 Per-reflection phase distributions

Inverting (5) via Newton–Raphson gives the concentration:

$$\hat{X}_h = \left[I_1/I_0\right]^{-1}(\hat{m}_h) \tag{24}$$

The full phase probability distribution per reflection is then:

$$P(\varphi_h) = \frac{1}{2\pi\, I_0(\hat{X}_h)} \exp\left[\hat{X}_h \cos(\varphi_h - \varphi_{\text{mod}})\right] \tag{25}$$

This is the full distribution — not just a point estimate — enabling:

- Credible interval computation: $P(|\varphi_h - \varphi_{\text{mod}}| < \delta)$
- Phase-posterior sampling for conformational ensembles
- Information-theoretic measures (entropy, mutual information)

No existing software (REFMAC, Phenix, PanDDA, qFit, polder) provides this distributional output.

9.6 Alpha/beta shell estimation

Following Lunin & Skovoroda, the parameters α_k, β_k per resolution shell are recovered from the ensemble FOMs via:

$$\hat{X}_h = \hat{m}_h^{-1}(\hat{m}_h) \tag{26}$$

$$\hat{\gamma}_k = \frac{\sum_{h\in k} w_h\, b_h\, \hat{X}_h}{\sum_{h\in k} w_h\, b_h^2}, \quad b_h = \frac{|F_{\text{obs}}||F_{\text{mod}}|}{\varepsilon_h} \tag{27}$$

$$\hat{\beta}_k = \frac{\sum_{h\in k} w_h\, |F_{\text{obs}}|^2/\varepsilon_h}{\sum_{h\in k} w_h} \tag{28}$$

$$\hat{\alpha}_k = \tfrac{1}{2}\hat{\gamma}_k\, \hat{\beta}_k \tag{29}$$

where $w_h = 1$ (acentric) or $1/2$ (centric) and ε_h is the reflection multiplicity. The stochastic ensemble plays the role of the R-free test set, preventing overfitting bias.

10 Low-Occupancy Feature Detection

10.1 The phase-bias suppression problem

For a feature at occupancy q absent from the model:

$$F_{\text{obs}}(h) = F_{\text{true}}^{\text{protein}}(h) + q\, F^{\text{feature}}(h) + \text{noise} \tag{30}$$

The mean map uses model phases φ_{mod} derived from the feature-free protein model. The density at the feature site is attenuated by both the FOM and the phase shift δ_h caused by the feature:

$$\langle \rho_{\text{ens}} \rangle\big|_{\text{site}} \approx q \cdot m_h \cdot \cos(\delta_h) \cdot \rho_{\text{feature}}^{\text{full}} \tag{31}$$

For $m_h = 0.7$ and $\delta_h \sim 20$ at $q = 0.1$: total attenuation $\approx 0.66 \times 0.1 = 0.066$ — barely above noise.

10.2 The stochastic difference density

Proposition 10.1: Difference density linearity

The stochastic difference density:

$$\Delta\rho(\mathbf{r}) = \rho_{\text{obs}}(\mathbf{r}) - \rho_{\text{ens}}(\mathbf{r}) \tag{32}$$

has signal-to-noise at the feature site that scales linearly in q with no FOM attenuation factor:

$$\text{SNR}\big[\Delta\rho\big]\big|_{\text{site}} \approx \frac{q\,|F^{\text{feature}}|}{\sigma_{\text{noise}}} \tag{33}$$

Argument. By the mean-map identity: $\rho_{\text{ens}} \approx \mathcal{F}^{-1}[m_h|F_{\text{obs}}|e^{i\varphi_{\text{mod}}}]$, which is the MMSE estimate of the ground-state density. Subtracting from $\rho_{\text{obs}} = \mathcal{F}^{-1}[|F_{\text{obs}}|e^{i}]$:

$$\begin{aligned}\Delta F(h) &= |F_{\text{obs}}|e^{i} - m_h|F_{\text{obs}}|e^{i\varphi_{\text{mod}}} \\ &\approx (1 - m_h e^{i(\varphi_{\text{mod}}-)})|F_{\text{obs}}|e^{i}\end{aligned} \tag{34}$$

The phase shift $-\varphi_{\text{mod}}$ is dominated by the feature contribution, so ΔF carries the feature's Fourier components with amplitude proportional to $q|F^{\text{feature}}|$. The subtraction of $m_h|F_{\text{obs}}|e^{i\varphi_{\text{mod}}}$ removes the protein contribution to first order, and the residual noise σ_{noise} comes from the ensemble variance. The SNR thus scales as q/σ_{noise}, with no m_h attenuation. □ □

Comparison with equation (31): the mean map attenuates the feature signal by $m_h \cos(\delta_h) \ll 1$ at low occupancy; the difference density does not. This is the fundamental advantage of the difference approach and the direct analogue of PanDDA's multi-dataset subtraction.

10.3 Background Dataset Contribution

Following PanDDA, define the BDC as the fraction α such that:

$$\rho_{\text{event}} = \rho_{\text{obs}} - (1-\alpha)\rho_{\text{GS}} \tag{35}$$

minimises the correlation between ρ_{event} and ρ_{GS} in the non-event region. For the stochastic framework, $\rho_{\text{GS}} = \rho_{\text{ens}}$, and the BDC can be estimated as $\hat{\alpha} \approx 2(\text{positive fraction of } \Delta\rho - 0.5)$, where a positive fraction of exactly 0.5 corresponds to pure noise with no residual ground-state contribution.

11 Philosophical Discussion: What Is a FOM Without True Phases?

11.1 The epistemological status of FOM

The figure of merit m_h is conventionally described as the expected cosine of the phase error, suggesting it is a property of the true structure. This is misleading. The FOM is a property of *your probability distribution* over phases — it measures how concentrated your beliefs are, not how close you are to the truth.

Two crystallographers who begin with different models will compute different FOMs for the same dataset. Both are *correct* given their respective information states. There is no single objectively true FOM for a reflection, any more than there is a single objectively true probability for a coin toss.

11.2 Calibration as the only meaningful criterion

The question "is my FOM correct?" is therefore ill-posed unless it means: *is my stated FOM calibrated?* — i.e., do the phases I call 90% confident actually contain the true phase 90% of the time, across many reflections?

Calibration can be tested empirically when independent phase information is available (SAD/SIR phasing, ultrahigh-resolution Wilson statistics). The Lunin & Skovoroda (1995) demonstration is a calibration test: they show that LB estimates are calibrated while working-set Sigma-A estimates are not.

11.3 Three estimators, one quantity

The three FOM estimators in the literature all target the same underlying quantity — the concentration of the phase posterior — but from different directions:

1. **Sigma-A (REFMAC/Phenix):** likelihood maximisation over working reflections. Fast, available, but biased for refined models.

2. **LB estimates (Lunin):** likelihood maximisation over R-free reflections. Unbiased, but requires R-free partitioning and provides only shell-average estimates.

3. **Stochastic ensemble (EDENN):** empirical sampling of the phase distribution via atom omission. Unbiased (omit perturbations are not conditioned on any reflection subset), and additionally provides per-voxel spatial resolution via the SNR map.

11.4 What the identity with Blow-Crick means philosophically

The identity $\rho_{\text{ens}} = \rho_{\text{best}}$ does not mean the ensemble is redundant. It means the ensemble is *correctly calibrated*: its mean map is as good as it is theoretically possible to be with the available phase information. The ensemble then adds everything the mean map cannot provide: the distribution around that optimal estimate.

This is the same relationship as maximum likelihood estimation to Bayesian posterior inference. The MLE (the mean) is optimal in the point-estimate sense; the full posterior (the distribution) is what enables uncertainty quantification, propagation, and decision-making under uncertainty.

12 Comparative Framework

12.1 Method capabilities matrix

Capability	*Blow-Crick*	*Sigma-A*	*Lunin LB*	*PanDDA*	*polder*	*EDENN*
Best map (MMSE)	✓	✓	✓	–	–	✓
Unbiased FOM	–	–	✓	–	–	✓
Per-reflection FOM	✓	✓	–	–	–	✓
Full phase distribution	–	–	–	–	–	✓
Calibrated CI on φ_h	–	–	–	–	–	✓
Real-space SNR	–	–	–	–	–	✓
Phase-posterior sampling	–	–	–	–	–	✓
Single-dataset use	✓	✓	✓	–	✓	✓
Low-occ. events	–	–	–	✓	partial	✓
No model required	–	–	–	partial	–	–
Formal uncertainty	–	–	–	–	–	✓

Table 1: Capability matrix across principal methods. ✓ = full support; partial = limited to specific contexts; – = not provided.

12.2 Sensitivity at low occupancy

Method	SNR at site	Min. detectable q	Phase bias
Standard 2Fo-Fc	$q \cdot \lvert F_{\text{mod}}\rvert/\sigma$ (model biased)	∼0.40	Severe
Composite OMIT	$q \cdot m_h \cdot \cos\delta_h \cdot S/\sigma$	∼0.25	Reduced

Method	SNR at site	Min. detectable q	Phase bias
EDENN mean map	$q \cdot m_h \cdot \cos\delta_h \cdot S/\sigma$	∼0.25	Reduced
polder	$q \cdot S/\sigma$ (near surface)	∼0.15	Minimal
EDENN diff. density	$q \cdot \lvert F^{\text{feat}}\rvert/\sigma_{\text{ens}}$	∼0.05–0.10	None
PanDDA (N datasets)	$q \cdot \lvert F^{\text{feat}}\rvert/(\sigma/\sqrt{N})$	∼0.02–0.05	None

Table 2: Approximate SNR scaling and minimum detectable occupancy. S = resolution-dependent structure factor amplitude.

13 Benchmark Proofs for Each Proposition

13.1 Benchmark architecture

Each benchmark proof below has four components:

1. **Claim:** the formal proposition to be tested
2. **Ground truth:** what source of true information is used
3. **Metric:** the quantity measured
4. **Comparison:** which existing method provides the baseline

Benchmark Proof 13.1: P1 — Mean map identity with Blow-Crick

Claim: $\rho_{\text{ens}}(\mathbf{r}) = \rho_{\text{best}}(\mathbf{r})$ in the limit $K \to \infty$.
Ground truth: REFMAC5 Blow-Crick map (FWT column from REFMAC refinement on the same model and data). REFMAC provides the established gold standard for the Blow-Crick formula.
Metric: (i) $R_{\text{map}} = \sum_{\mathbf{r}} |\rho_A - \rho_B| / \sum_{\mathbf{r}} |\rho_A + \rho_B|$ between ensemble mean and REFMAC map, plotted as a function of K. Expected: $R_{\text{map}} \to 0$ as $K^{-1/2}$. (ii) Pearson correlation coefficient between $|\langle \tilde{F}_h^{\text{omit}} \rangle| / |F_{\text{obs}}|$ and REFMAC's output FOM column, per resolution shell. Expected: $r > 0.98$.
Comparison: Against Phenix composite OMIT map (FWT column). Composite OMIT *does not* converge to Blow-Crick; showing that your ensemble does, while composite OMIT diverges, establishes a formal distinction between the methods.
Requirement for passing: $R_{\text{map}} < 0.01$ at $K = 20$ on a test PDB structure (suggested: 1CRN or equivalent small protein with published REFMAC outputs).

Benchmark Proof 13.2: P2 — Unbiased FOM estimation

Claim: $\hat{m}_h$ from the stochastic ensemble is unbiased relative to Sigma-A estimates, specifically for overfitted models.
Ground truth: SAD-phased experimental phases φ_{true} (available for your 5FIO dataset), which allow direct computation of the empirical FOM: $m_h^{\text{emp}} = \cos(\varphi_{\text{mod}}(h) - \varphi_{\text{true}}(h))$.
Metric: Per-shell mean of $|m_h^{\text{emp}} - \hat{m}_h|$ for: (a) EDENN stochastic estimate, (b) REFMAC Sigma-A, (c) Lunin LB estimate. Plot as a function of resolution d^{-1} — the Lunin & Skovoroda Figure 1.
Comparison: The expected outcome is that EDENN and LB estimates track m_h^{emp} while REFMAC Sigma-A overestimates FOM at medium to high resolution (a known systematic

bias documented in the original Lunin & Skovoroda paper).
Requirement for passing: EDENN shell-average bias $|\hat{m} - m^{\text{emp}}|$ must be smaller than REFMAC shell-average bias in at least 70% of shells between 3 Å and 1.5 Å.

Benchmark Proof 13.3: P3 — Phase distribution calibration

Claim: The von Mises phase distribution $P(\varphi_h)$ with concentration $\hat{X}_h$ is calibrated: the nominal γ% credible interval contains φ_{true} in γ% of reflections.
Ground truth: SAD experimental phases φ_{true} (5FIO).
Metric: Empirical coverage at $\gamma \in \{68\%, 90\%, 95\%\}$ for acentric reflections, and the analogous Bernoulli test for centric reflections. Coverage curves plotted against resolution.
Comparison: REFMAC's FOM column does not provide credible intervals — there is no existing method to compare against. The comparison is therefore against the theoretical nominal level (the $y = x$ line on a calibration plot).
Requirement for passing: Empirical coverage within ±5% of nominal for $d > 2.0$ Å. Above 2.0 Å coverage may decrease due to overfitting of high-resolution shells.

Benchmark Proof 13.4: P4 — SNR as evidence score for water molecules

Claim: The EDENN SNR at a modelled water site predicts whether that water is a genuine solvent molecule or a noise artifact.
Ground truth: Water survival analysis: refine the structure with all waters included, then remove all waters and re-run automated water placement (Phenix or ARP/wARP) starting from a randomised model. Waters that are re-placed in the second run are classified as "genuine"; those that are not re-placed are classified as "artifacts".
Metric: Area under the ROC curve (AUC) for predicting genuine waters from EDENN SNR. Compare with: (a) B-factor of the water in the original model, (b) peak height in the 2Fo-Fc map, (c) real-space correlation coefficient from Phenix.
Comparison: On the BAZ2B dataset (your Aim 1 data), using the re-placement classification as labels. Expected: EDENN SNR has higher AUC than B-factor alone because B-factor conflates coordinate error with genuine disorder, while SNR measures density evidence directly.
Requirement for passing: AUC ≥ 0.80 for EDENN SNR; AUC of EDENN must exceed AUC of B-factor by ≥ 0.05.

Benchmark Proof 13.5: P5 — Difference density low-occupancy sensitivity

Claim: The stochastic difference density (32) detects low-occupancy ligands that are invisible in the Blow-Crick mean map.
Ground truth: XChem BAZ2B screening campaign PanDDA event labels (available from XChem XCDB), which provide ground truth for which crystals contain bound fragment and at what approximate occupancy.
Metric: For each single dataset independently: (i) Z-score of $\Delta\rho$ at the PanDDA-defined event centroid, (ii) Z-score of ρ_{ens} (mean map) at the same centroid. Plot both Z-scores against the PanDDA-assigned occupancy estimate.
Comparison: Against polder map Z-score at the same centroid (polder maps are the current best-practice single-dataset approach). Expected: $Z[\Delta\rho]$ is higher than $Z[\rho_{\text{ens}}]$ at all occupancies, and comparable to polder at occupancies > 0.15 while superior at occupancies < 0.15.
Requirement for passing: For datasets where PanDDA-assigned occupancy $q < 0.20$,

$Z[\Delta\rho] > 3.0$ in $\geq 70\%$ of cases while $Z[\rho_{\text{ens}}] < 3.0$ in the same cases.

Benchmark Proof 13.6: P6 — Single-dataset event recovery vs. PanDDA

Claim: On the BAZ2B campaign, the stochastic difference density recovers ligand-binding events detected by PanDDA, from single datasets, with sensitivity comparable to PanDDA's campaign-level estimate.
Ground truth: PanDDA event list for BAZ2B (available from Pearce et al. 2017 supplementary data and XChem XCDB).
Metric: Precision and recall at Z-threshold 3.0 for: (a) EDENN stochastic difference density (single dataset), (b) PanDDA event detection on the same single dataset (PanDDA applied to $N = 1$, which is an invalid use case — should give near-zero sensitivity), (c) PanDDA on the full campaign ($N \sim 100$). The expected ordering is: EDENN single-dataset > PanDDA single-dataset, EDENN single-dataset < PanDDA full campaign.
Requirement for passing: EDENN recall exceeds PanDDA recall at $N = 1$ by ≥ 0.3 across all occupancy bins; EDENN precision > 0.6.

Benchmark Proof 13.7: P7 — K-convergence rate

Claim: FOM estimation error decreases as $O(K^{-1/2})$ consistent with the central limit theorem for the ensemble average.
Ground truth: Large-K limit ($K = 200$) FOM estimate as reference, or SAD experimental phases where available.
Metric: Mean absolute error of $\hat{m}_h$ vs. K for $K \in \{3, 5, 10, 20, 50, 100\}$. Fit to $\text{MAE}(K) = a/\sqrt{K} + b$ and report the R^2 of the fit and the extrapolated asymptotic bias b.
Comparison: Against composite OMIT map, which has no formal convergence theory — composite OMIT does not improve systematically with the number of iterations, because each sub-volume is independently simulated.
Requirement for passing: $R^2 > 0.95$ for the $1/\sqrt{K}$ fit; asymptotic bias $b < 0.03$ (residual systematic error at large K is negligible).

Benchmark Proof 13.8: P8 — Alpha/beta recovery vs. Lunin LB

Claim: The stochastic $\hat{\alpha}_k$, $\hat{\beta}_k$ estimates recover the Lunin LB parameters without requiring explicit R-free partitioning.
Ground truth: Lunin LB estimates computed from the actual R-free set of the test structures (using CCP4's SFCHECK or a custom implementation of the LB likelihood).
Metric: Per-shell correlation and mean absolute error between EDENN $\hat{\alpha}_k$ and Lunin α_k^{LB}, plotted against resolution.
Comparison: Against REFMAC working-set Sigma-A estimates, which are known to be biased. Expected: EDENN and Lunin LB agree with each other and both diverge from REFMAC Sigma-A in the 2–1.5 Å resolution range.
Requirement for passing: Pearson $r > 0.90$ between EDENN and LB estimates across shells; EDENN MAE relative to LB must be smaller than REFMAC MAE relative to LB in $\geq 60\%$ of shells.

14 The Complete Value Proposition

14.1 What Blow-Crick cannot provide

The Blow-Crick formula requires a complete, well-refined model. Its FOM estimates are biased for refined models (the working-set problem). It provides no uncertainty information beyond the single scalar m_h. It has no direct application to detecting unmodelled features.

14.2 What EDENN adds

Restated precisely: given a model, EDENN.waters provides:

1. **The same best map as Blow-Crick**, but with unbiased FOM estimates (because omit perturbations are not conditioned on any reflection subset).
2. **The full von Mises phase distribution** $P(\varphi_h)$ per reflection, enabling calibrated credible intervals — available from no other method.
3. **The real-space SNR map** $\mathrm{SNR}(\mathbf{r})$, providing a continuous measure of density evidence at every voxel — available from no other method.
4. **A stochastic difference density** $\Delta\rho$ whose SNR at unmodelled sites scales as q/σ_{ens} with no phase attenuation — analogous to PanDDA events but operative on a single dataset.
5. **Phase-posterior sampling**, enabling the conformational ensemble required by EDENN.autobuild to correctly propagate measurement uncertainty into structural heterogeneity models.

14.3 Final statement

Blow-Crick maps are the *mean* of the EDENN framework. EDENN is the *distribution* around that mean. The distribution is what enables detection, calibration, propagation, and decision-making under uncertainty that no point-estimate framework — Blow-Crick, Sigma-A, composite OMIT, or polder — can provide.

A Mathematical Appendix

A.1 Von Mises characteristic function

For $\varphi \sim \mathrm{vonMises}(\mu, \kappa)$:

$$\mathbb{E}[e^{in\varphi}] = \frac{I_{|n|}(\kappa)}{I_0(\kappa)}\, e^{in\mu}, \quad n \in \mathbb{Z} \tag{36}$$

At $n = 1$: $\mathbb{E}[e^{i\varphi}] = (I_1(\kappa)/I_0(\kappa))\, e^{i\mu} = m_h\, e^{i\mu}$.

A.2 Expected phase error integral

The mean absolute phase error for acentric reflections:

$$\langle|\Delta\varphi|\rangle = \frac{\int_0^\pi |\Delta\varphi|\, e^{X_h \cos\Delta\varphi}\, d(\Delta\varphi)}{\pi\, I_0(X_h)} \tag{37}$$

Evaluated numerically by Gauss–Legendre quadrature on $[0, \pi]$.

A.3 Centric reflections

For centric reflections, $\varphi_h \in \{0, \pi\}$ only (Friedel mates are symmetry-related). The phase probability is Bernoulli:

$$P(\varphi_h = \varphi_{\mathrm{mod}}) = \frac{e^{X_h/2}}{2\cosh(X_h/2)}, \qquad P(\varphi_h = \varphi_{\mathrm{mod}} + \pi) = \frac{e^{-X_h/2}}{2\cosh(X_h/2)} \tag{38}$$

The FOM for centrics is $m_h = \tanh(X_h/2)$.

A.4 Circular variance

The circular variance of $\cos(\varphi_h - \varphi_{\text{mod}})$ under the von Mises distribution:

$$\text{Var}[\cos \Delta\varphi] = \tfrac{1}{2}(1 + I_2(X_h)/I_0(X_h)) - m_h^2 \tag{39}$$

For large X_h: Var $\approx (1 - m_h^2)/2$.

A.5 Wilson B-factor and the Debye-Waller factor

The Debye-Waller factor D_h in equation (6) is equivalent to an additional B-factor: $D_h = \exp(-B_{\text{coord}}\, s^2/4)$ where $B_{\text{coord}} = 8\pi^2\sigma_x^2$. This connects the coordinate-error model directly to B-factor analysis and explains why EDENN's FOM estimates should be compared to Wilson B-statistics in validation.

A Principal References

Blow & Crick (1959)
The treatment of errors in the isomorphous replacement method. *Acta Crystallogr.* **12**, 794–802. *Derivation of the best Fourier map and the figure of merit.*

Read (1986)
Improved Fourier coefficients for maps using phases from partial structures with errors. *Acta Crystallogr. A* **42**, 140–149. *Sigma-A formalism; D and Sigma parameters; connection to Luzzati.*

Brünger (1992)
Free R value: a novel statistical quantity for assessing the accuracy of crystal structures. *Nature* **355**, 472–475. *R-free test set; motivation for unbiased estimation.*

Lunin & Skovoroda (1995)
R-free likelihood-based estimates of errors for phases calculated from atomic models. *Acta Crystallogr. A* **51**, 880–887. *LB estimates; unbiased alpha/beta from R-free set; Figure 1.*

Terwilliger et al. (2008)
Iterative model building, structure refinement and density modification with the PHENIX AutoBuild wizard. *Acta Crystallogr. D* **64**, 61–69. *Composite OMIT maps; simulated-annealing debiasing.*

van den Bedem et al. (2009)
Exploiting single-crystal diffraction data to model the conformational ensemble of proteins. *Methods Enzymol.* **474**, 357–376. *qFit; multiconformer model building from density.*

Liebschner et al. (2017)
Polder maps: improving OMIT maps by excluding bulk solvent. *Acta Crystallogr. D* **73**, 148–157. *Polder maps; bulk-solvent exclusion.*

Pearce et al. (2017)
Partial-occupancy ligands in protein crystallographic fragment screens. *Nat. Commun.* **8**, 15816. *PanDDA algorithm; event maps; BDC; fragment screening.*

Riley et al. (2021)
qFit 3: Protein and ligand multiconformer modeling for X-ray crystallographic and single-particle cryo-EM density maps. *Protein Sci.* **30**, 270–285. *qFit 3; linear program formulation; occupancy estimation.*